When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

A local speech model can keep a conversation going, and a stronger expert can take over the hard questions. Calling the expert on every turn adds cost and waiting time, so the local model needs an early estimate of whether its own answer will fail. Hidden-state probes give such estimates for text models. It is not known whether this signal survives speech and duplex fine-tuning, audio input, and the timing of a native listen/speak protocol. We study these questions on frozen MiniCPM-o 4.5. We use lightweight probes to read failure signals from hidden states and separately calibrate a gate for expert handoff at answer onset. We find that failure signals can transfer from text to speech, and that readout design matters in native audio generation. On our internal benchmark, selective expert handoff raises answer-content accuracy from 40.8% to 62.9%. In an independent real-time replay, server waiting time after input end to first answer audio has P50 0.70 s locally and 2.87 s with aggressive handoff. These timing sessions have not been scored for correctness, so the two experiments do not establish joint accuracy–latency performance.

1 Introduction

Spoken interaction leaves little time to choose a response [Stivers et al., 2009]. One practical design pairs a local speech model that keeps the conversation going with a stronger expert for hard questions. Calling the expert on every turn adds cost and waiting time. Selective escalation requires an estimate of local-answer failure risk that is available when the model begins to respond.

Prior work gives two starting points. For text models, hidden-state probes and verbal self-evaluation can predict whether an answer is correct [Kadavath et al., 2022, Azaria and Mitchell, 2023, Kossen et al., 2024], and intermediate layers can carry information that the output does not show [Orgad et al., 2025]. Text routers use such signals to divide queries between models of different cost [Chen et al., 2023, Ong et al., 2024, Varshney et al., 2026]. For speech, native duplex models decide for themselves when to listen and when to speak [Défossez et al., 2024, Cui et al., 2026].

Three things remain unknown. First, it is not known whether a failure probe still works when the query pool changes; a high in-distribution AUC can come from pool identity rather than from competence. Second, it is not known whether a probe fit on text input still reads speech-input states after speech and duplex fine-tuning. Third, it is not known where in a streaming protocol the hidden state is actually available for a routing decision. A duplex model already decides when to speak, but this floor-management decision need not reveal whether it knows the answer. Although the pretrained backbone can carry competence information, its accessibility across layers, read positions, and input modalities remains unclear in native streaming speech after duplex fine-tuning.

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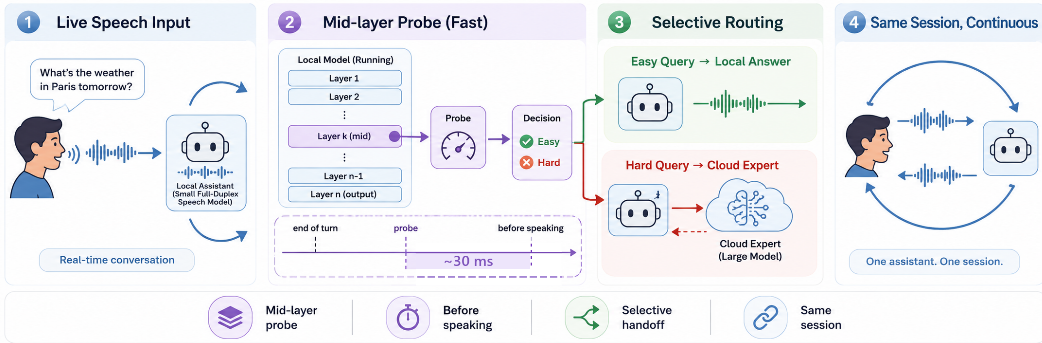

Figure 1: **The deployed system.** **(1)** Speech streams in 1 s chunks into a full-duplex local model that decides for itself when to speak. **(2)** At that speak commit, a mid-network (L22) linear read scores the turn in $\sim$30 ms at answer onset; the triggering call may already return the first text unit. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert while the talker stalls. **(4)** The returned answer is spoken verbatim in the talker's own voice. Delivered accuracy on the internal test set rises from 40.8% to 62.9% at 52% escalation in native full-duplex sessions.

This gap matters in practice. A gate placed at the wrong layer can look good on its calibration pool and then invert on a new one, so the expert is called on the wrong turns. A probe that fails on audio input cannot be used in a speech system at all. And a read that only becomes available after the answer is generated removes the early decision that motivated escalation in the first place.

We study these questions on frozen MiniCPM-o 4.5 [Cui et al., 2026], with raw-backbone and other speech-model controls. We make three contributions. First, a layer and position audit shows that the accessibility of failure information depends on the input and calibration regime, motivating our L22 read (Figure 2). A matched native ablation separates the contribution of feature aggregation from layer choice (Figure 4). Second, text-trained probes transfer to speech-input states, reaching 0.760 AUC on query-disjoint human recordings (Figure 5). Cross-model controls test the breadth of this transfer, which remains distinct from the calibration needed for native serving. Third, a separately fitted answer-onset gate raises internal answer-content accuracy by 22.1 percentage points and exceeds the reported random-mixture reference by 6.8 points at the aggressive tier. An independent real-time replay measures server waiting time after input end to first answer audio. These unjudged timing sessions do not establish joint accuracy–latency gains or client-audible delay.

2 Related Work

Competence readout. Correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024], and uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]. Intermediate-layer representations can expose information missing from model outputs [Orgad et al., 2025, Mahaut et al., 2024]; Semantic Entropy Probes amortize uncertainty estimation into a hidden-state read [Kossen et al., 2024]. Our question is how readout location, input modality, and serving protocol affect this signal after speech and duplex fine-tuning.

Selective routing. FrugalGPT, HybridLLM, and RouteLLM allocate work across language models [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024]; AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Pre-generation activation routers and frozen-state success probes are especially close to our method [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear failure probe is therefore not itself novel. We study its robustness in speech representations and calibrate thresholds for a tunable expert-call budget; local failure remains distinct from expert benefit.

Native duplex control. Native spoken-dialogue models coordinate listening and speaking [Défossez et al., 2024, Cui et al., 2026], using state or synchronization mechanisms [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024] or frozen-backbone adapters [Wang et al., 2024b]. BayLing-Duplex [Fang et al., 2026] makes dialogue-state decisions within a single autoregressive LLM by interleaving user audio, assistant text, and assistant audio and introducing four state tokens. Starting from GLM-4-Voice, it learns turn-taking and interruption through supervised full-duplex fine-tuning and timing-focused preference optimization. Its evaluation measures spoken-answer quality and interaction timing, providing a concrete example of learned native floor control.

Competence-aware handoff. DuplexOmni, chronological thinking, and MoshiRAG connect speech to reasoning or retrieval [Huang et al., 2026, Wu et al., 2026, Chien et al., 2026]. Our work studies a decision complementary to native floor control: given a duplex-trained checkpoint, can its hidden states predict local-answer failure and guide escalation to an expert under a tunable call budget? We fit a small readout without updating the speech model. This requires regime-specific calibration and does not assume that a learned turn-taking signal predicts answer correctness.

3 Probe-Guided Expert Handoff

3.1 The local–expert loop

Selective model routing balances response quality against the cost of a stronger model [Ong et al., 2024]. This suits our goal: improve answer accuracy under a limited expert-call budget. We adapt the routing decision to a local full-duplex speech model's transition from listening to speaking (Figure 1). The local model processes incoming speech and decides when to respond; the gate then chooses whether to continue locally or request expert help. On the expert path, an audio uplink transcribes the question, the expert returns a text answer, and the local talker voices the prepared relay text. The request and relay add cost and waiting time. We use local-answer failure risk as a proxy for escalation value; gains depend on expert correction and faithful relay of the answer.

3.2 A probe for the handoff decision

Linear *probes* test what information a classifier can read from frozen representations [Alain and Bengio, 2016]. Hidden-state classifiers have estimated statement truthfulness [Azaria and Mitchell, 2023] and pre-generation task success [Lugoloobi et al., 2026]. We use this approach to ask whether a speech model's existing states contain a directly readable failure signal. Reusing these states avoids generating a separate self-evaluation and leaves the speech model unchanged. The linear restriction tests directly accessible information without ruling out information requiring a nonlinear readout.

Offline, we collect local answers to calibration questions and use a judge to assign correctness labels. Each label is paired with hidden-state features captured at the chosen read point. We fit a logistic probe to predict failure from these features while keeping the speech model frozen. For features $\phi(x)$ from input context x, the fitted weights w, b give the risk score

$$s(x) = \sigma\big(w^\top \phi(x) + b\big)\,. \tag{1}$$

At inference, the probe reads the current features without a reference answer or an online judge. Scores above a language-specific threshold τ_ℓ favor escalation; thresholds select nominal expert-call budgets from calibration scores and stay fixed during evaluation (§4.1). Realized call rates can change when the query distribution shifts. An auxiliary dialogue-act head restricts escalation to information requests, filtering turns such as acknowledgments and stop commands (Appendix D).

3.3 Where should the probe read?

Adapting the probe to duplex speech requires choosing its layer, position, and read time. The final layer feeds the output head, but its failure signal can degrade under a change in query distribution or input modality. Figure 2 motivates L22 through text-input leave-one-pool-out (LOPO) evaluation: on held-out math, its last-token probe reaches 0.931 AUC, versus 0.372 at the final layer. This failure does not persist when both calibration and inference use audio input (§4.4).

The native feature vector concatenates the latest L22 tail state, the mean of the tail states, and the mean input state, allowing the probe to combine the current boundary state with pooled context. The matched native ablation in §4.4 shows that this aggregation matters: a single L22 state does not outperform a single final-layer state. Both depth and aggregation therefore define the native readout.

Figure 2: **Text-input transfer depends on layer and position.** LOPO math AUC across layers: late last-token reads deteriorate on the duplex checkpoints, while mid-layer reads and raw-backbone controls remain predictive. Layer 22 is the MiniCPM-o 4.5 read location used for the native gate. This plot does not show an equivalent collapse under audio calibration and audio input.

We read these features when the generation call first returns a listen-to-speak transition. This makes the gate available at the local model's response boundary without waiting for a completed answer. The call can already contain the first generated text unit, so we call the decision point *answer onset*. Further answer chunks may expose additional failure information, at the cost of a later handoff (Appendix D.2). We fit the native probe and thresholds on features and answer labels from this serving regime to match its audio input and generation behavior. This calibration is separate from the representation-transfer study (§4.5), which alone does not provide a zero-shot native gate.

4 Experiments

4.1 Experimental setup

We evaluate frozen MiniCPM-o 4.5 (9B), with `gpt-5.5` as the escalation expert, on the fixed 240-query internal test set and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], and SD-QA from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. For the answer-content benchmark, the native probe is fitted on 5,221 calibration rows after excluding benchmark overlaps. Its results were rerun with this refitted probe; per-language thresholds use the remaining core rows' out-of-fold scores and stay fixed across pools. Calibration details and model controls are in Appendix A.

Each query has one sampled native session per arm: local-only, conservative, balanced, aggressive, and always-escalate. We report *answer-content accuracy*, scored from generated local text or intended relay text. The random reference is the analytical mixture $A_{\mathrm{mix}}(r) = (1-r)A_{\mathrm{local}} + rA_{\mathrm{always}}$ at the gate's realized call rate r. Table 1 uses the original query weights; Figure 3's Internal curve uses the category weights specified in its caption. Both accuracy views use the same recorded outcomes. External averages weight the four QA pools equally. In this content benchmark, the expert receives the final 30 seconds of the prerecorded question. Timing comes from an independent `ttfa-v3` run on the same query IDs, using real-time input, only the consumed audio prefix for expert requests, audio generation on both paths, and prepared relay text. Server TTFA measures the wait from scheduled input end until the first answer PCM block is ready; audio ready earlier incurs zero wait. Candidate and stall audio are excluded; client playback is unmeasured. These timing-session answers have not been judged, so accuracy and TTFA are not joint measurements. Appendix A.1 provides the model, probe-fitting, and answer-judging configurations used in these experiments.

4.2 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports answer-content accuracy on the fixed 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive. Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix D.1). Aggressive exceeds the

Table 1: **Main results: answer-content accuracy (%).** Both blocks report the internal test set and four external English speech-QA pools; avg is the equal-pool external mean and Δ is its gain over always-local, computed before rounding. MiniCPM uses one recorded native session per query and arm, with fixed calibration thresholds. Matched random mixes its local and measured always arms at each pool's realized gate rate (Appendix D.1). NVDA uses native-frame replay with cached outcomes; its Internal column is a post-selection diagnostic. Replay details are in Appendix C.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	avg	Δ
MiniCPM native sessions (answer-content accuracy)							
always-local	42.3	62.8	52.8	82.4	48.0	61.5	—
+ gate, conservative	50.1	60.4	59.2	78.8	54.0	63.1	+1.6
+ gate, balanced	63.6	70.0	62.4	82.4	65.0	70.0	+8.5
+ gate, aggressive	**72.6**	**88.4**	**75.2**	**85.6**	**82.5**	**82.9**	**+21.4**
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	77.1	+15.6
always-escalate (measured)	77.6	96.0	79.6	91.6	87.0	88.6	+27.1
NVDA three-layer answer-onset probe (pass-3 native-frame replay)							
always-local	19.6	35.8	37.6	70.8	31.0	43.8	—
+ gate @15%	31.7	48.4	48.8	80.7	42.0	55.0	+11.2
+ gate @30%	44.6	61.0	59.2	85.8	54.0	65.0	+21.2
+ gate @50%	60.4	76.8	69.6	89.7	66.5	75.7	+31.9
matched random @50%	53.5	65.7	60.2	82.0	60.0	67.0	+23.2
always-expert (ceiling)	91.7	95.5	82.8	93.1	89.0	90.1	+46.3

table's random reference of 56.1% by 6.8 percentage points. Per-tier internal comparisons are in Table 9. Aggressive recovers 74.6% of the measured local-to-always accuracy gap while using 51.7% expert calls. This measures selection value relative to the recorded endpoints.

Gains vary across the fixed internal strata: factual QA improves by 52.5 points, while math accuracy is unchanged. Of 60 knowledge questions, 43 exceed the expert's input window. These descriptive results show task differences and an input constraint; they do not isolate a truncation effect on accuracy or change the reported test mixture and denominators (Appendix A.2).

The independent `ttfa-v3` run measures server waiting time to first answer audio after input end (Figure 3). Internal P50 is 0.70 s locally, 1.61 s at balanced, and 2.87 s at aggressive. Aggressive has mean 6.10 s and P95 18.14 s; always has mean 9.43 s and P50 6.67 s. Local mean is 1.06 s, including 17/240 attempts (7.1%) that respond before input end and incur zero wait. Early responses do not establish better interaction or correctness. Table 10 counts failures and reports timing among completed sessions. These answers remain unjudged: 62.9% accuracy and 6.10 s mean TTFA are results from separate experiments and do not establish a joint accuracy–latency operating point.

4.3 External transfer and its limits

The native gate's weights, feature definition, layer, and per-language thresholds stay fixed across the four external English QA pools. The upper block's *avg* excludes the internal set; Δ is its gain over always-local. Aggressive macro accuracy rises from **61.5% to 82.9%**, compared with 77.1% for the reported random mixture: +21.4 points over local answering and +5.9 over random. Its realized call rate ranges from 18.8% on Llama Questions to 66.0% on SD-QA, so this is not a uniform 50%-budget comparison. TriviaQA and SD-QA illustrate large improvements over local answering (+25.6 and +34.5 points) and over the corresponding random reference (+6.9 and +8.8 points).

The lower tiers are less reliable. Conservative accuracy falls by 2.4 points on TriviaQA and 3.6 on Llama Questions; balanced leaves Llama Questions unchanged. Separately sampled sessions and limited expert headroom constrain the interpretation of small differences. The results support aggressive-tier gains across these pools, not improvement on every benchmark at every budget. On AlpacaEval, we measure 3.68 locally and 4.01 at aggressive, with a random-mixture reference of approximately 3.9; the always arm scores 4.20 versus 4.95 for returned expert text. The length cap on relayed answers is particularly restrictive for tasks requiring long, open-ended responses.

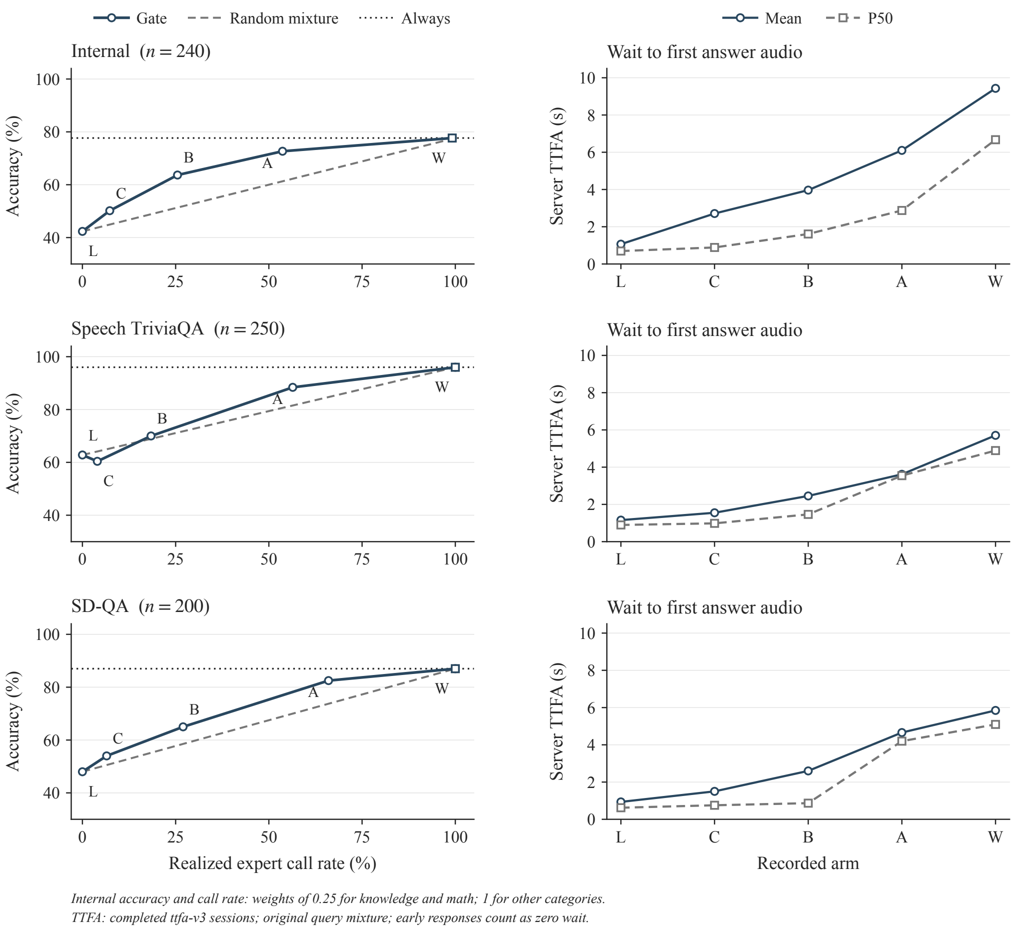

Figure 3: **Accuracy and independently measured TTFA on three pools.** Left: answer-content accuracy against expert call rate. C/B/A denote conservative/balanced/aggressive. The dashed line follows the table's random-mixture convention (§4.1); the horizontal reference is the measured always arm, not an oracle bound. Right: mean and P50 server TTFA from completed sessions in the independent `ttfa-v3` run, with original query weights. TTFA is the nonnegative wait after input end until first answer PCM; early answers have zero wait. L/W denote local/always. Each query has one attempt per arm; candidate/stall audio and client playback are excluded. Timing-session answers are unjudged and do not establish the accuracy on the left. Internal counts and tails appear in Table 10; Table 1 covers all four external English speech-QA pools evaluated here.

Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC increases from 0.755 at commitment to 0.791 after two answer chunks. Later chunks expose useful failure information without identifying the mechanism underlying this change (Appendix D.2).

The lower block of Table 1 reports the exploratory NemotronLabs-VoiceChat-11B (NVDA) study. Its *avg* covers four English QA pools and excludes the Internal column. At a 50% replay budget, average accuracy rises from 43.8% to 75.7% ($\Delta = 31.9$ points), compared with 67.0% for random. This is native-frame replay with cached outcome mixtures, not another set of MiniCPM native sessions. The L26/L30/L34 heads reach 0.818 out-of-fold AUC and 0.816 mean external AUC in the separate probe analysis (Appendix C); Table 1 reports accuracy rather than AUC. External pools were inspected during variant review, so independent confirmation is needed.

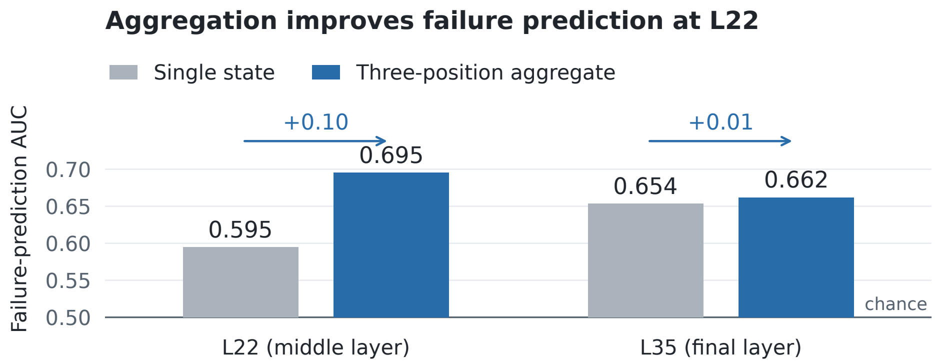

Figure 4: **Aggregation improves failure prediction at L22.** Mean AUC across five external pools under leave-one-pool-out evaluation on one native audio batch. Gray bars read the latest tail state; blue bars concatenate that state with the tail mean and input mean (§3.3). Arrows indicate the aggregation gain at each layer; the horizontal line marks chance AUC of 0.5.

4.4 Readout ablations

Text-input transfer. We evaluate the readout without using expert outcomes. The sweep in Figure 2 compares read positions and raw-backbone controls; mean pooling reduces late-layer degradation. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96, with similar ordering in the Qwen2.5 control family (Table 5). With audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. These controls link degradation to the tested speech checkpoints and input regimes without identifying its cause (Appendix B.2).

Native layer and aggregation. We test whether predictive gains at native answer onset come from layer choice or feature aggregation. A 2×2 comparison crosses L22 versus the final layer with a single tail state versus the three-part aggregate defined in §3.3. All four readouts use one audio batch with identical training pools, labels, splits, and head recipe (Figure 4). At L22, aggregation raises mean failure-prediction AUC from 0.595 to 0.695; at the final layer, the change is smaller (0.654 to 0.662). The single-state L22 read has no advantage over the final layer. These results identify an aggregation benefit at L22, without establishing an advantage from middle-layer selection alone. Named per-pool results and cached routing gains are reported in Appendix B.4.

4.5 Text-trained probes transfer to unseen human speech

We next test whether a probe fitted to text-input hidden states can predict failure under speech input. This tests a change of input modality; Figure 2 tests held-out task-pool transfer within text input.

Figure 5 (left) holds question content fixed: the same 600 queries are presented as text and as TTS audio. The text-trained MiniCPM-o 4.5 probe reaches **0.855** AUC on audio states at L22. Transfer also appears in MiniCPM-o 2.6 and Qwen2.5-Omni, with plateau AUCs of 0.74–0.80 and 0.80–0.83, respectively; the frozen-backbone Freeze-Omni control shows much weaker transfer under the tested recipe. This panel tests modality transfer with shared question content.

The right panel tests generalization to *new questions and human speech*: probes fitted on the calibration pool are evaluated on 200 query-disjoint SD-QA recordings. The text-trained MiniCPM-o 4.5 probe reaches **0.760** AUC at L22, with a per-layer peak of 0.800 at L25. Audio-trained probes and evaluation on SD-QA text provide references for the modality change. Thus, transfer extends beyond matched TTS questions to held-out human speech. The two panels use different evaluation sets, so their AUC difference does not isolate a human-speech penalty (Appendix B.1). These results concern representation transfer; the native gate requires a separately fitted readout (§3.3).

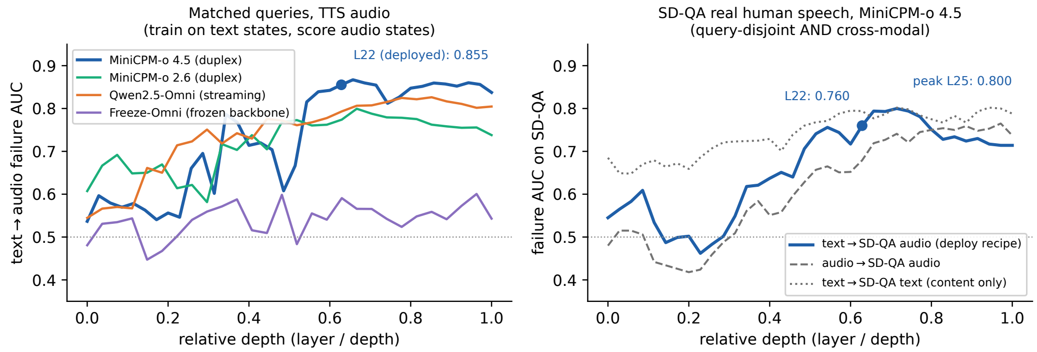

Figure 5: **Text-trained failure probes transfer to unseen human speech.** Left: text-trained probes are evaluated on TTS audio for the same 600 questions across four models; MiniCPM-o 4.5 reaches 0.855 AUC at L22. Right: MiniCPM-o 4.5 probes fitted on the calibration pool are evaluated on 200 query-disjoint SD-QA human recordings. The text-trained probe (blue) reaches 0.760 at L22 and peaks at 0.800 at L25; gray curves show audio-trained and text-target references. Both panels show relative depth, normalized by the total number of layers in each model.

5 Discussion and Limitations

Three boundaries define the scope of the results. First, coverage is limited to the tested checkpoints and predominantly single-turn speech QA. Broader conversational tasks, including multi-turn dialogue and expert continuity, remain to be evaluated. Second, native serving requires in-regime audio features and answer labels to fit the readout and set thresholds; representation transfer alone does not provide a zero-shot gate. Third, accuracy and server TTFA come from separate runs; timing-session correctness is unjudged. Early responses need not improve turn-taking. Human-rated speech quality and timestamps of audio playback on the client remain unmeasured in these experiments.

6 Conclusion

We find failure information in frozen speech models that transfers from text to unseen human speech. At native answer onset, an aggregated readout supports selective expert calls and improves answer-content accuracy without updating the speech model. These results connect representation analysis to a calibrated handoff decision, complementing the model's control of when to speak. Establishing conversational benefits requires joint evaluation of correctness, audible latency, and turn-taking.

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A Benchmark and calibration details

The main table reports the original internal query mixture and four external English QA pools. The figure's Internal accuracy and call-rate curve uses post-hoc knowledge/math weights of 0.25 and weights of 1 for other categories; timing retains the original mixture. Archived Mandarin diagnostics are reported separately and excluded from the main table's English-pool macro accuracy average.

A.1 Experimental setup details

Models and decoding. The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Matched controls include raw backbones, a second MiniCPM generation, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni (Table 3). Representation analyses use bf16, greedy decoding, and seed 42. Native decoding uses temperature $T = 0.7$ and top-$k = 20$. The expert is `gpt-5.5` with low reasoning effort.

Probe fitting and thresholds. Representation probes use 360 calibration queries and single-position 4,096-dimensional features. The native gate uses 5,221 calibration rows and three L22 summaries: the last tail state, the tail mean, and the input mean. Its logistic head uses $C = 3 \times 10^{-4}$. Five-fold stratified cross-validation uses shuffle seed 42. For each language, conservative, balanced, and aggressive thresholds are the $(1 - r)$ quantiles of the 4,979 core rows' out-of-fold scores, with $r \in \{0.15, 0.30, 0.50\}$. The 242 fresh training rows enter the fit but not the quantile base.

Judge configuration. Speech TriviaQA, Speech Web Questions, and Speech Llama Questions use the OpenAudioBench judge with `gpt-4o-2024-08-06`. Internal, SD-QA, and Mandarin reasoning answers use the reference-anchored `gpt-5.4-mini` adequacy judge. AlpacaEval uses VoiceBench's `gpt-4o-mini` judge on its 1–5 scale. The adequacy judge receives the question, an optional reference, and a candidate answer without the system identity. Its rubric requires factual correctness, valid reasoning, and a complete answer. Prompt details are in Appendix E.

A.2 Internal task strata

Table 2: Descriptive strata of the fixed internal test set. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm's realized rate. The final column counts question waveforms longer than the expert's 30-second input window. Each query contributes one recorded run per arm, with all samples retained at their original weights.

Stratum	n	Local	A	Always	A call rate	> 30 s (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The largest gains occur on factual QA and SimpleQA traps. Math is unchanged at the aggregate level, consistent with the failure-type analysis in Appendix D.2. Long-input truncation is concentrated in the knowledge stratum (43 of 60 questions); all queries remain in the reported denominator.

B Models and representation analyses

B.1 Signal comparison and layer transfer

The final-layer state predicts source pool at 95.8% accuracy, and a pool-identity oracle alone reaches failure AUC 0.715. This motivates LOPO rather than an in-distribution comparison. The depth sweep shows that the failure is localized: mid-layer last-token reads transfer while late reads degrade on both duplex models; mean pooling degrades less. In the signal table, the final-layer audio entry is audio-calibrated LOPO math, whereas L22 tests a text-calibrated probe on speech states.

The text-to-audio result is not explained by synthetic speech alone: on 200 human SD-QA recordings, the text-calibrated read remains at 0.76–0.80 AUC across L22–L26. Audio-input verbal self-evaluation is less reliable; transcripts restore it, while probing needs no extra generation.

Table 3: Model roster. “Pair” identifies a fine-tuned model and its raw backbone control.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control

Table 4: Failure-prediction AUC on MiniCPM-o 4.5. In-dist uses five-fold out-of-fold predictions; LOPO fits on other pools and evaluates on the named held-out query pool.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	near chance	—
$p(\text{True})$-pre	0.807	trap 0.945	degrades
$p(\text{True})$-post	**0.899**	—	degrades
final-layer probe	0.822	**0.372**	0.936
mid-layer (L22) probe	0.866	**0.931**	**0.855**

B.2 Mechanism checks

Matched controls narrow the late-layer effect without assigning it to a specific training objective. A speak-mode template leaves final-layer LOPO math unchanged (0.362 versus 0.366), the audio–text direction is nearly orthogonal to the failing probe, and source-pool identity is decodable at all depths. A residual-component analysis shows the distributed read holding through the middle layers and decaying only near the outputs of the two duplex models; it remains intact in their raw backbones and the omni-streaming control. These controls support a mid-layer read in the tested conditions, while leaving the causal role of individual fine-tuning objectives unresolved.

B.3 Query-surface baseline

A TF–IDF logistic router reaches 0.669 out-of-fold AUC and 0.38–0.57 under LOPO, including 0.230 on the all-failure trap pool. The hidden-state probe’s advantage is therefore not explained by query wording or source identity alone. On audio labels it improves to 0.814, below L22’s 0.879.

B.4 Native feature ablation

Figure 4 uses one native audio batch (`agg0`, official configuration), with all layer features captured in the same sessions and judged against their own generated answers. Single-state and aggregated features have 4,096 and 12,288 dimensions, respectively. Each readout fits a logistic head ($C = 3 \times 10^{-4}$) on four pools and evaluates the fifth. The five pools include Mandarin reasoning; this set differs from the main table’s four English external pools. Table 6 identifies the individual pool results that underlie the averages, including the weaker reasoning scores.

The routing comparison selects the highest-risk 30% of questions in each pool and combines recorded local outcomes with cached always-arm expert outcomes. Figure 6 reports pooled accuracy gains over 2,000 matched random allocations. Per-pool gains in Table 6 use the analytical 30% random mixture. These are cached outcome comparisons, not new live gate sessions. For the L22 single-state read, 0.0505 of the random allocations matched or exceeded the observed accuracy; none did for the other three readouts. These tests compare each readout to random routing only.

For context, the separately trained deployed gate reaches mean AUC 0.750 and a pooled gain of 4.80 percentage points on these rows. It was fitted on the full calibration merge, so it is not a matched training condition for the four LOPO ablations and is excluded from their plots.

Table 5: LOPO math AUC by depth for matched model families.

model	best mid-layer	final layer	shape
Qwen3-8B	0.964 (L33/36)	0.958	plateau
MiniCPM-o 4.5	**0.931 (L22/36)**	**0.366**	late inversion
Qwen2.5-7B	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni	0.794 (L16/28)	0.746	diffuse decline
MiniCPM-o 2.6	**0.822 (L21/28)**	**0.540**	late cliff

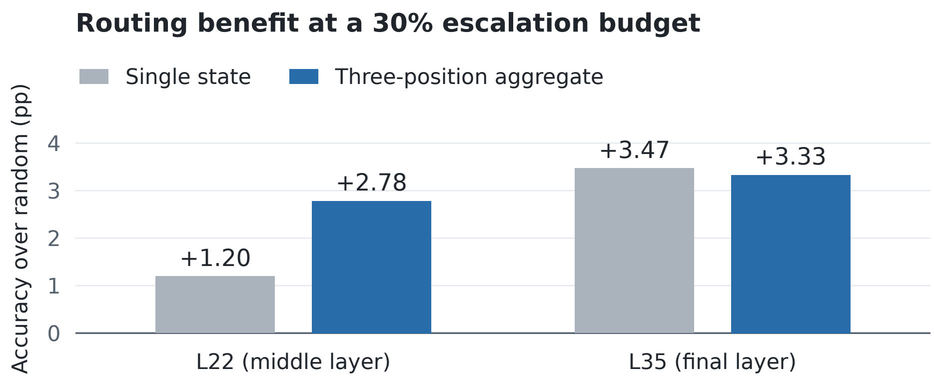

Figure 6: **Cached routing gains at a matched 30% budget.** Accuracy gains over random escalation, pooled across the same five evaluation sets as Figure 4. Each bar uses the recorded local and expert outcomes; feature choices and colors match the main ablation figure.

C Second duplex family

We replay NemotronLabs-VoiceChat-11B through its native frame protocol with the checkpoint frozen. Of 2,550 English calibration queries, 2,481 produce a commit frame. Standardized logistic heads read L26/L30/L34 onset states; their normalized logits are averaged. The recipe is chosen on calibration-only LOPO metrics, and external labels are excluded from fitting. Each head uses $C = 10^{-4}$ and concatenates the commit state, eighth post-commit state, first-eight-state mean, and running pre-commit mean (17,920 dimensions per layer). External pools were inspected during variant review, so these results provide exploratory transfer evidence. The read covers the first 640 ms after commit, so this experiment tests method transfer rather than end-to-end real-time serving.

The three-layer read reaches 0.818 out-of-fold AUC and 0.816 mean external AUC under the pool-aligned judges. A strictly causal commit-only variant loses about 0.03 external AUC, making answer onset the closest matching read point across the two tested model families.

D Native gate and serving details

The L22 hidden state is available after roughly 57–61% of prefill. On one H100, probe availability has P50/P95 of 20/25 ms for text and 45/104 ms for audio. These numbers measure the local read, not the network or expert call. The native benchmark's realized rates are below; their variation is expected because thresholds are calibration quantiles rather than per-pool quotas.

Dialogue-act filtering. A second linear head uses the same L22 read to distinguish information requests from floor-management turns. Escalation requires both the information-seeking score $a(x)$ and failure score $s(x)$ to exceed their respective thresholds in the same turn:

$$d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\}\, \mathbf{1}\{a(x) \geq \tau_a\}. \tag{2}$$

On the scripted calibration stimuli, the act head has OOF AUC 1.000; the question-side 0.5th-percentile threshold removes 0.5% of true escalations and admits no floor-management stimuli.

Table 6: Native ablation by evaluation pool. Each column is one of the four matched LOPO readouts in Figure 4. The lower block gives accuracy gains over random escalation in percentage points. AUC averages weight pools equally; pooled routing gains weight individual questions.

	L22		L35	
Pool	Single	Aggregate	Single	Aggregate
Failure-prediction AUC				
Speech TriviaQA	0.645	0.744	0.704	0.661
Speech Web Questions	0.579	0.732	0.646	0.683
Speech Llama Questions	0.614	0.711	0.672	0.674
SD-QA	0.648	0.713	0.682	0.669
Reasoning QA (zh)	0.488	0.576	0.564	0.624
Equal-pool mean	0.595	0.695	0.654	0.662
Accuracy gain over random at a 30% budget (pp)				
Speech TriviaQA	+3.38	+4.58	+3.78	+3.38
Speech Web Questions	+1.69	+1.69	+4.16	+3.33
Speech Llama Questions	+1.51	+3.96	+2.73	+3.96
SD-QA	+2.85	+5.35	+5.85	+4.35
Reasoning QA (zh)	-3.91	-1.93	+1.04	+1.53
Pooled gain	+1.20	+2.78	+3.47	+3.33

Table 7: VoiceChat-11B transfer. Accuracy equally weights the four external English speech-QA pools; the random reference uses the same escalation rate as the probe-guided policy.

expert budget	15%	30%	50%
probe-guided accuracy	55.0	65.0	75.7
random-mixture accuracy	50.7	57.7	67.0

D.1 Realized escalation rates

Table 8: Realized escalation rates (%) for the MiniCPM native results in the upper block of Table 1, plus the archived Mandarin and separate AlpacaEval pools. These are measured rates, distinct from the nominal calibration budgets and from the NVDA replay budgets. Local-only never escalates.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

Table 9: Internal native-protocol answer-content accuracy with query-bootstrap 95% confidence intervals ($n = 240$). Random follows Table 1's endpoint-mixture convention.

tier	realized calls	accuracy	random reference
local-only	0.0%	.408 [.35,.47]	.408
conservative	8.3%	.467 [.40,.53]	.433
balanced	25.0%	.542 [.48,.60]	.482
aggressive	51.7%	.629 [.57,.69]	.561
always	99.2%	.704 [.65,.76]	.704

Table 10: Internal server TTFA (seconds), independent `ttfa-v3` run: 240 attempts per arm, with 239/238/236/236/237 completed and 1/2/4/4/3 failed in row order. TTFA is the nonnegative wait from scheduled input end to first answer PCM. All completed sessions count; the 17/12/11/13/7 early responses have zero wait. Failed attempts have no TTFA value. Timing ends at the first local PCM-bearing chunk return or relay PCM yield, excluding candidate/stall audio and client playback.

tier	mean	P50	P95	P99
local-only	1.06	0.70	2.62	3.64
conservative	2.71	0.89	10.85	34.08
balanced	3.97	1.61	14.41	26.91
aggressive	6.10	2.87	18.14	54.78
always	9.43	6.67	25.86	60.83

D.2 Failure type and later reads

This separate sweep fits a probe at each read point using 5,236 judged calibration rows, row-random five-fold fitting, and $C = 3 \times 10^{-4}$. Later reads pad finished answers with the last available vector. Its five-pool macro includes Mandarin reasoning, unlike the main table's four-English-pool macro. Failure categories are assigned by a model judge from recorded answers.

Failure-prediction AUC at successive native read points. External macro equally weights five speech-QA pools.

Read point	Calibration OOF	Internal	Reasoning-zh	External macro
Before onset	.848	.832	.711	.753
At onset	.850	.852	.648	.755
After 1 chunk	.852	.854	.681	.762
After 2 chunks	.855	.846	.769	.791
After 3 chunks	.857	.858	.744	.801

Specific-but-wrong answers account for about 70% of failures and are ranked at AUC 0.813; perception errors reach 0.889, while execution slips are near chance at 0.469. The later-read gains on reasoning expose a tradeoff between earlier routing and information that emerges during generation.

E Judge prompt and reproducibility

The answer judge receives the question, an optional reference, and a candidate answer without the system identity. It marks the candidate adequate only when it is factually correct, uses valid reasoning, and completely answers the question; partial, hedged, refused, off-topic, and truncated answers are inadequate. The pre-answer self-evaluation asks the model not to answer the question and to reply exactly “Yes” or “No” to whether it would answer correctly. The post-draft version presents the question and proposed answer and asks whether the answer is correct.

Representation-study sampling and splits use seed 42. The native benchmark uses sampled decoding and retains one run per query and arm. MiniCPM-o 4.5 runs with PyTorch 2.8.0 and Transformers 4.51.0 on single-H100 instances. The release includes the calibration manifest, out-of-fold scores, thresholds, overlap exclusions, benchmark hashes, and scripts that rebuild the figures and accuracy and timing summaries from the original recorded data for these experiments.